Erfan Nasirzadeh Orang
Machine Learning · Data Analytics · Human-Centered Computing

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Education

Master of Computing and Information Systems (MCIS) Aug 2026 – Present
Youngstown State University, OH
Advisor: Dr. Zack While · Thesis Based
Fully Funded Graduate Research Assistantship

Computer Science, Bachelor of Science with Honors Aug 2022 – May 2026
Youngstown State University, OH
Minors: Mathematics, Data Analytics · Honors Scholar with Distinction
GPA: 3.50/4.00 cumulative, 3.73/4.00 major
Dean's List (4 semesters) · President's List (2 semesters)

Research Experience

Graduate Research Assistant, Computer Science Dept., YSU
Advisor: Dr. Zack While May 2026–Present

Project: HCI/fMRI Neural Encoding

- Evaluated whether a neural encoding model (Meta's TRIBE v2) can replicate human fMRI results from a visualization perception study, matching model-predicted cortical responses to human behavioral data in visual-processing regions.
  Skills *PyTorch, Python, replication methodology, HPC cluster*

> **E. N. Orang** and Z. While. *Can a Neural Encoding Model Replicate an fMRI Visualization Study?* VISxVISION Workshop, IEEE VIS '26 (2026). [arXiv:2609.15685]

Project: Physics-Informed ML Surrogates

- Trained ML surrogate models (XGBoost, Random Forest, Gaussian Process Regression) on CFD and thermal-system data for wind turbine and battery applications, using SHAP and nested cross-validation to extract spatially resolved uncertainty estimates.
  Skills *Python, scikit-learn, XGBoost, GPR, SHAP, nested cross-validation*

> A. Ulus, I. Uluocak, **E. N. Orang**, N. Vakili, K. Choo. *ML Surrogates for Al_2O_3–TiO_2/Water Hybrid Nanofluid Battery Thermal Management.* JTAC, 2026 (Accepted).

> A. Ulus, I. Uluocak, **E. N. Orang**, N. Vakili, K. Choo. *Turbulence effects on an H-type Darrieus vertical-axis wind turbine using uncertainty-aware surrogate modeling.* Part A: J. Power and Energy (Under Review, 2026).

Project: Embedded Interactive Information Displays (EIID) — in preparation

- Collaborating with a Ph.D.'s student on qualitative analysis of embedded interactive information displays, performing collaborative qualitative coding.
  Skills *Qualitative analysis, Condens*

Lab Research Assistant, Depts. of Physics & Biology, YSU Oct 2022–May 2025

Project: Quantum Optics Lab — Zeeman Electromagnetically Induced Transparency

- Conducted laser-based quantum optics experiments and performed theoretical modeling and signal processing.
  Skills *Python, MATLAB, instrumentation, calibration*

> J. Gorkos, K. Grenzig, **E. N. Orang**, V. Thomas, D. Tighe, M. Crescimanno. *Chirp Asymmetry in Zeeman Electromagnetically Induced Transparency.* JOSA B, 42(2), 404–411 (2025).

Project: Imaging Lab Internship — Photonic Nanostructures

- Trained and certified on AFM, SEM, and TEM; collaborated with Florida Polytechnic University on optical measurements of photonic nanostructures.

SKILLS *Nanoscale imaging, precision instrumentation*

> A. Fiorito, D. R. Sheffield, H. Liu, **E. N. Orang**, N. J. Dawson. *Spectral fingerprint of laser emission from rhodamine 6g infused male Indian peafowl tail feathers.* Scientific Reports, 15, 20938 (2025).

Oral/Poster Presentations

Spectral fingerprint of laser emission from rhodamine 6g infused male Indian Peafowl tail feathers. Oral presentation by **E. N. Orang**, Fall 2025 Joint Meeting of the Eastern Great Lakes Section and MIAAPT, Ypsilanti, Michigan, October 2025.
[*Link to Session Abstract*]

A Mathematical Model of Foot Traffic-Induced Wear on Historical Staircases: A Solution to the COMAP Mathematical Contest in Modeling Problem. Oral presentation by **E. N. Orang**. Pi Mu Epsilon Honor Society Annual Conference, Youngstown State University, March 2025.
[*Link to Session Abstract*]

Chirp Asymmetry in Zeeman Electromagnetically Induced Transparency. Poster presentation by **E. N. Orang** at the 59th Annual Conference of the National Collegiate Honors Council (NCHC), San Diego, California, September 2025.
[*Link to Session Abstract*]

Chirp Asymmetry in Zeeman Electromagnetically Induced Transparency. Poster presentation by **E. N. Orang** at the 55th Annual Meeting of the APS Division of Atomic, Molecular, and Optical Physics (DAMOP), Session S00.38, Fort Worth, Texas, June 2024.
[*Link to Session Abstract*]

Press Coverage

"Peacock feathers can be lasers", Science Magazine (AAAS), July 2025.[*Link to Article*]

"Study Detects Lasers in Peacock Tail Feathers", The Business Journal, August 2025.
[*Link to Article*]

"Florida Poly student research uncovers laser light from peacock feathers", Florida Polytechnic University News, August 2025.[*Link to Article*]